Bridging the Digital Divide in Public Health: A Comparative Evaluation of Retrieval-Augmented Generation for Universal Health Coverage Intelligence in LMIC Nations

BACKGROUND

Low-middle-income countries face persistent digital health divides impeding UHC attainment under Sustainable Development Goal (SDG) 3.8. WHO Health Policy Systems Research (HPSR) identifies need for quantitative performance assessment with qualitative governance frameworks to generate actionable health intelligence.[1] ".Yet RAG deployment in resource-constrained settings raises governance challenges of reliability, equity, and algorithmic accountability.

OBJECTIVE

To evaluate open-source RAG platforms across reliability, equity, and governance dimensions for UHC policy intelligence in LMICs, and to propose an empirically grounded governance framework

METHODS

A pragmatic mixed-methods embedded design evaluated two platforms (AnythingLLM’s; RAG Flow), across two language models (Llama 3.2 3B; Gemma 3 27B) under two parameter configurations (permissive vs. restrictive) using WHO SDG documents (2005-2025). We assessed 200 query-responses for factual accuracy, hallucination rates, protocol adherence, and response consistency (Fleiss' κ=0.82) using WHO ground truth benchmarks to conceptualize RAG-PHI governance framework using context-coding.[2]

RESULTS

Restrictive configurations reduced hallucination by 62.5% for RAG Flow with Llama 3.2 3B (Cohen's d = 0.89; $F[1,7992] = 384.2$, $p < 0.001$; $\eta^2 = 0.046$), achieving highest factual accuracy ($M = 2.31$, $SD = 0.67$; ICC = 0.97). Critical failures included Gemma 3 27B generating fabricated India's UHC Service Coverage Index (88.9 vs. WHO-verified 54.3) across both configurations, RAG Flow non-response to 85% of UHC queries under permissive settings and AnythingLLM's response latency variance of 30-fold (500 ms to 15 s) for identical queries.

DISCUSSIONS

Configuration insensitivity in Gemma 3 27B reveals a training-corpus governance failure unresolvable by parameter tuning, generating cross-layer insights unavailable from either quantitative or qualitative analysis independently.

CONCLUSION

Current RAG implementations requires substantial improvement before responsible LMIC deployment utilizing proposed conceptualized RAG-PHI framework (ethics-as-infrastructure) operationalized empirically across three layers: knowledge-based sovereignty, configuration accountability, deployment equity. Future research should evaluate multilingual capabilities, SMS/voice modalities for deployment across diverse LMIC health systems.

KEYWORDS

Digital health equity; Health information systems; Low- and middle-income countries; Public health informatics; Retrieval-Augmented Generation; Universal Health Coverage

1.0 INTRODUCTION

Global pursuit of Universal Health Coverage under SDG 3.8 confronts a fundamental paradox: two decades of WHO reporting (2005-2025) generates vast health datasets, yet administrators who most urgently require these insights for policy formulation often cannot access them. A district health officer in rural Bihar managing over 600 public healthcare centres may spend 24 - 48 hours manually aggregating data across repositories before answering a fundamental question as to how many children received measles vaccines in their district over three months implying not a data scarcity problem but a health intelligence access problem.

WHO HPSR 'Changing Mindsets' strategy advocates health research as the GPS of health decision-making, requiring embedded navigation tools to ascertain health problem, identify desired outcome, mapping feasible routes to improvement.[1] LMICs face structural barriers that impede this navigation: fragmented information architectures, HR shortages, limited digital infrastructure, and linguistic diversity that renders most AI health tools inaccessible to non-English-speaking populations.[3–5]

Retrieval-Augmented Generation (RAG) represents a promising approach to bridging this divide by grounding language model responses in verified external documents rather than relying solely on pre-trained knowledge. A well configured RAG system retrieves relevant passages from a curated document collection before generating responses unlike standard chatbots, which may hallucinate plausible-sounding but factually incorrect information.[6] There are very few sparse studies on systematic evaluations of RAG platform reliability for public health intelligence in the context of LMIC nations. This study addresses three research objectives is framed around a pragmatic embedded mixed-methods design:

RQ1: How do platform, language model, and parameter configuration affect factual accuracy, hallucination rate, protocol adherence, and response consistency in open-source RAG systems evaluated against WHO UHC ground-truth data?

RQ2: What empirically-derived governance framework based on the failure-mode patterns of RAGFLOW and Anything LLM can responsibly guide RAG deployment for public health intelligence across LMIC settings?

RQ3: How do quantitative configuration-performance findings, when systematically integrated with implementation science frameworks, generate governance insights that neither methodological approaches could independently generate?

These objectives converge on a central governance proposition: RAG configuration choices are infrastructure-level governance decisions with direct equity consequences for LMIC health administrators.

2.0 LITERATURE REVIEW

2.1 Healthcare AI and RAG Systems

Recent studies on RAG in healthcare highlights the potential of grounding large language models (LLMs) in curated evidence bases to mitigate hallucinations in clinical and biomedical contexts.[7–8] RAG-powered health intelligence systems must serve dual purposes in LMIC contexts: support high-precision policy analysis (ex: 'How has out-of-pocket health expenditure changed across LMIC nations between 2015-2023?') and provide accessible citizen guidance on health policies, facilities, and schemes.[9–10]

Traditional decision support systems excel at pre-structured queries against defined data warehouses but fail to meet ad-hoc analytical demands.[11–12] RAG addresses this limitation by grounding LLM responses with verified external information[6] which is critical in healthcare where information hallucinates imposes direct risks upon patient safety and public health decision-making.[7]

2.2 Governance Frameworks for AI in Public Health:

WHO HPSR strategy establishes that health systems are highly complex, multi-layered, and nonlinear, requiring research designs combining quantitative and qualitative methods to assess interventions.[1] HPSR strategy argues that health policy and systems research must transcend individual discipline boundaries and function as trans-disciplinary, demand-driven inquiry embedded within decision-making ecosystems.

Rogers et al.'s integrative context coding framework, integrated with Consolidated Framework for Implementation Research (CFIR), offers a systematic approach to evaluate multi-level contextual determinants impacting implementation outcomes across public healthcare systems.[2] The framework consolidates organizational documents, interview data, and observational records into a structured mechanism that maps contextual influences across individual, team, organizational, and system levels. This study applies analogous multi-level reasoning to RAG governance, positioning configuration choices, knowledge base architecture, and deployment modality as contextual determinants of implementation equity and not solely as incidental technical parameters. The existing global frameworks offer only partial coverage as explicitly stated in WHO (2021) framework along with OECD AI Principles which establishes useful normative benchmarks but does not address RAG-specific governance across resource-constrained healthcare deployments across LMIC nations.[13].

2.3 Research Gaps

A PRISMA-guided search across (PubMed n=182, Embase n=121, PsycInfo n=32) identified 335 studies; 20 met inclusion criteria after de-duplication, abstract based title screening (shared in the supplementary file of GitHub repository). Four substantive gaps emerged: (i) a persistent clinical-versus-policy imbalance, with 89% of LLM studies concentrated on clinical applications and none evaluating RAG for SDG monitoring or UHC tracking;[7] (ii) reliance on single-platform designs, sparse or none cross-platform comparative evaluations of open-source systems under identical conditions;[14–15] (iii) acknowledgment of parameter effects without systematic quantification;[16–17] and (iv) LMIC-oriented conceptual roadmaps that lacks grounding in empirical deployment evidence.[18]

3.0 THEORETICAL AND CONCEPTUAL FRAMEWORK

This study grounds its governance framework in two complementary theoretical foundations: the WHO HPSR strategy's argument that health systems research must embed evidence generation within decision-making ecosystems,[1] and context coding framework stated in Rogers et al.'s establishes that implementation success depends on systematic multi-level assessment of contextual determinants.[2] Both frameworks opines that governance must be embedded in data architectures , configuration regimes from inception and not retrofitted as downstream compliance, but must be entrenched within data architectures, configuration regimes and institutional arrangements from inception. When applied to RAG-based health systems, this means that infrastructure choices governs similarity thresholds, temperature settings, retrieval depth, and embedding strategies collectively determine which queries receive reliable responses and whose health realities are rendered legible within the system and which population it is catered to. This study present findings to substantiate that the hallucinations observed were not random model errors but predictable consequences of design and training choices.

3.1 The RAG-PHI Conceptual Framework

The literature review provides a theoretical foundation in this study to conceptualize and propose Retrieval-Augmented Generation for Public Health Intelligence (RAG-PHI) framework as its primary conceptual contribution. RAG-PHI articulates three interlocking governance layers derived empirically from the factorial experiment's failure-mode analysis:

Layer 1: Knowledge-Base Sovereignty

The corpus is treated as a governance artefact, not a neutral repository. Document selection, metadata enrichment, chunking strategy, and quality control shape representational equity. In LMIC deployments, national and sub-national data - including district-level KPIs and indigenous health knowledge - must be prioritized alongside global WHO reports.

Layer 2: Configuration Accountability

Parameters such as temperature, similarity threshold, top-K retrieval, and penalty terms are governance levers that must be documented, justified, and verifiable. For high-stakes policy queries, restrictive configurations that favor precision and explicit 'I don't know' responses over fluency are normatively required.

Layer 3: Deployment Equity

Responsible deployment demands multilingual interfaces, infrastructure-appropriate modalities (SMS and voice for low-bandwidth settings), transparent uncertainty communication, and phased roll-out in which human-verified support precedes partially automated decision-support. RAG-PHI aligns with WHO HPSR strategy's call for governance frameworks that are 'embedded within decision-making processes' with Rogers et al. framework's emphasis on monitoring contextual dynamics longitudinally across system levels shown in **Figure 1**: RAG-PHI Conceptual Framework (visualized by author using AI-assisted image generation tool, saved as Tiff. File)

Figure 1 Alt Text: A conceptual diagram of the RAG-PHI governance framework structured as three nested layers. The innermost layer represents Knowledge-Base Sovereignty, the intermediate layer represents Configuration Accountability, and the outermost layer represents Deployment Equity. Directional arrows trace the influence pathway from independent governance pillars through mediating process variables toward dependent outcomes, specifically hallucination reduction and factual accuracy.

3.2 Theoretical Framework Variables

RAG-PHI governance framework integrates three theoretical domains through four independent pillars influencing system efficacy via three mediating process variables. Contextual factors moderate these pathways, optimizing four dependent outcomes while controlling for corpus size, model architecture, and prompt structure. (Tables 1 and 2 operationalize the framework variables and is shared in the supplementary file as Table_S1: Supplementary S1 and Table_S2 as Table_S2: Supplementary S2) whereas Figure 2 shows theoretical framework for the complete operationalization of the RAG-PHI framework variables wherein construct definitions mapped, System Configuration and Parameters of the models as shown in Figure 2. saved as Tiff. file.

Figure 1 Proposed Conceptual Theoretical framework using RAG PHI, System Config and Parameters of RAG PHI

Figure 2 Alt Text: Theoretical framework diagram mapping RAG-PHI construct types such as ***Independent, Mediating, Moderating, Dependent, and Control variables*** which are interconnected boxes with directional arrows showing hypothesized causal pathways from knowledge sovereignty with configuration accountability accounting to health policy intelligence outcomes.

Design Element	Quantitative Component	Qualitative/ Conceptual Component
Purpose	Evaluate RAG platform reliability for UHC policy intelligence	Develop RAG-PHI governance framework grounded in empirical findings

Design Element	Quantitative Component	Qualitative/ Conceptual Component
Approach	2×2×2 factorial experiment (platform × model × configuration)	Deductive framework synthesis using WHO HPSR and implementation science
Data source	WHO SDG documents (2005-2025); Total = 200 query-response pairs	Failure-mode analysis; Rogers et al. CFIR-informed context coding
Analysis	ANOVA, effect sizes (η^2, Cohen's d), ICC, Fleiss' κ=0.82	Thematic mapping of failure modes to RAG-PHI governance layers
Ground truth	WHO UHC SCI benchmarks (Ex: India=54.3)	WHO DHSPR 'Changing Mindsets' framework; Rogers et al. context coding
Integration point	Quantitative failure modes inform qualitative framework layers	RAG-PHI layers operationalize experimental findings as governance levers
Ethical approval	Exempt - no human subjects; secondary analysis of public WHO data	N/A since it is a conceptual framework in development

Table 1*:* Methodology Summary (JARS-Mixed Items 4a-4r*)*

Table 1 Alt Text: "Summary table of mixed-methods design elements comparing Purpose, Approach, Data source, Analysis method, Ground truth, Integration point, Ethical approval across the Quantitative Component (Phase 1) and Qualitative/Conceptual Component (Phase 2) columns."

4.0 METHODOLOGY

A brief Methodology Summary as per JARS Mixed Items Checklist is detailed in Table 1.

Note: Integration strategy: Quantitative failure modes from Phase 1 were systematically mapped to RAG-PHI governance layers in Phase 2, following integration design recommended for embedded mixed-methods studies.

4.1 Research Design

This study espouses a pragmatic mixed-methods embedded design in which quantitative experimental evaluation (Phase 1) directly apprises qualitative governance framework development (Phase 2). The mixed-methods is used to justify the study's dual purpose: Phase 1 contributes towards causal evidence about configuration-hallucination relationships, while Phase 2 translates technical findings into actionable governance recommendations for LMIC health administrators and policymakers. Complex health system challenges require methodological pluralism that neither quantitative nor qualitative designs can fulfil alone (WHO HPSR strategy).[1]

A critical-realist epistemology underpins the evaluation: verifiable ground-truth values exist for many queries (e.g., India's UHC Service Coverage Index in 2022 = 54.3), while constructs such as 'hallucination' and 'protocol adherence' necessarily involve expert interpretive judgment. The evaluation framework therefore goes beyond standard information retrieval metrics to incorporate dimensions with direct equity relevance, including protocol adherence, response consistency, and transparency in failure-mode identification. [1]

Figure 3: Unified System Architecture depicting RAG Flow and AnythingLLM's agent pipelines, showing document processing, vector-based retrieval, and response generation. Parameter positions for Configuration A and Configuration B are shown at respective retrieval and generation stages. (Visualized by author using AI-assisted image generation tool and saved as TIFF file]

Figure 2 Alt Text: A flowchart represents two parallel retrieval-augmented generation pipelines side by side. The left pipeline represents AnythingLLM's, which employs Chroma DB for vector retrieval. The right pipeline represents RAG Flow, which uses FAISS-based hybrid retrieval. Both pipelines has stages: document ingestion, text chunking, embedding generation, vector search, and language model response generation. The positions at which Configuration A and Configuration B parameters take effect is labelled at the retrieval and generation stages of each pipeline.

The integration strategy based on concurrent triangulation protocol adapted for embedded mixed-methods designs following O'Cathain et al. (2008). Quantitative outputs from Phase 1 shows specifically failure-mode frequencies, Cohen's d effect sizes, and per-condition ICC values were treated as structured inputs to the Phase 2 framework coding process. Each identified failure mode was then rated against Rogers et al.'s multi-level contextual determinant taxonomy, using a three-point scale (negative contextual factor = −2; neutral = 0; enabling factor = +2). This produced a documented analytical pathway connecting observed statistical anomalies to their corresponding governance implications. This approach fulfills the requirements of JARS-Mixed Item 4r and grounds the RAG-PHI framework in experimental evidence rather than solely-based upon theoretical assumption.

4.2 Experimental Design (JARS Items 4p-4q) :

The study employed a 2×2×2 factorial design organized around three factors: (1) platform (AnythingLLM's versus RAG Flow); (2) language model (Llama 3.2 3B versus Gemma 3 27B, with Tiny Llama on AnythingLLM's included as a robustness condition); and (3) parameter configuration (permissive Config A, replicating default out-of-box settings, versus restrictive Config B, which operationalizes *ethics-as-infrastructure principles*) illustrated in Figure 3 saved as TIFF file. Twenty standardized queries spans four categories: numerical factual, temporal trends, categorical lists, and explanatory prompts each administered 20 times per condition under a Latin-square randomization protocol, yielding 200 query-response pairs distributed across eight experimental conditions (*see Table S4_b for ICC values across 20 iterations*). Llama 3.2 (3B) was selected as a compact model suited to edge deployment, while Gemma 3 (27B) operates at a substantially larger parameter scale. This disparity in model size is treated as a contextual interpretive variable rather than a controlled factor, since differences in parameter count may contribute atleast in part to divergent configuration sensitivity observed between the two models (*Table S3, Experimental Configuration, supplementary file*).

4.2a Standardization and Confound Controls

Four standardization controls were applied uniformly across all experimental conditions to protect internal validity. Identical system prompts were used within each platform (see Appendix 1). Query administration followed a Latin-square randomization sequence to guard against order effects. All evaluations were carried out on the same institutional hardware with no recourse to external API calls (*see Supplementary Reproducibility Statement, Table S8).* Finally, inter-rater calibration was conducted on 20 pilot query-response pairs prior to main coding, so that the four-point factual accuracy scale was interpreted consistently across all coders and conditions (Fleiss' $\kappa = 0.82$ at main coding).

4.3 Knowledge-Based Construction (JARS Items 4c-4f)

The knowledge base comprised publicly available WHO SDG monitoring reports, UHC publications, and technical guidance documents spanning 2005–2025, yielding 21,629 overlapping segments chunked at approximately 1,000 tokens with 20% overlap. Different default embedding models were used across platforms (Anything LLM: all-MiniLM-L6-v2, tiny llama; RAG Flow: Nomic-embed-text), constituting a platform-native confound acknowledged as a study limitation in Section 6.2 to evaluate retrieval and response generation independently. Large documents (>1 GB) were parsed, cleaned of tables and figures, and chunked into tokens prior to testing. Document pre-processing included removal of embedded figures, tables, and statistical annexes prior to chunking to reduce noise in semantic retrieval. Metadata tags (document type, year, issuing body) were applied to all 80 source documents to enable future retrieval filtering. The resulting corpus of 21,629 segments represents the primary knowledge-based governance artefact under RAG-PHI Layer 1 (Knowledge-Base Sovereignty); its composition directly determines which health realities are rendered legible to the system and currently missing in governance decision with equity implications for LMIC users.

4.4 Outcome Measures and Coding (APA JARS Items 4g, 4h, 4k-4m)

Three independent coders with public-health training assessed each response on four dimensions:

i. Factual accuracy (exact/partial/incorrect/fabricated, scored 0-3);

ii. Hallucination presence (binary);

iii. Protocol adherence (appropriate 'I don't know' response when the knowledge base lacked sufficient evidence); and,

iv. Consistency (Intraclass correlation across 20 repetitions per condition) shown in Table S4_b: ICC using 20 iterations.

Inter-rater agreement showed Fleiss' κ=0.82. Ground-truth values for numerical queries were drawn from WHO UHC Service Coverage Index reports and SDG 3.8 monitoring data. Response time was recorded as a secondary operational measure, defined as "*the elapsed duration in seconds from query submission to full response display*" to evaluate deployment feasibility across frontline LMIC settings where latency has a direct bearing on practical usability. Most responses were mostly hallucinated in the context of health query for instance, the appearance of financial data was treated as a qualitatively distinct failure mode, separate from binary hallucination classification, coded independently in the analytic trail (see Supplementary Table S5, Row 4). This coding decision was established a priori and recorded in the inter-rater calibration protocol before any main coding occurred.

4.5 Ethics and Data Governance (APA JARS Items 4j, 4n, 4o)

This study doesn’t fall under the purview of institutional review board oversight, since it doesn’t involve human participants and draws exclusively on publicly available WHO SDG monitoring documents. No personally identifiable information was collected or processed at any stage. All RAG platforms were run on local institutional hardware using open-source models, so that no data were transmitted to external servers. The use of AI-assisted tools for generating Figures 1 and 3 is disclosed in accordance with JAMIA's AI usage policy.

4.6 Reproducibility and Open Science:

All Docker configuration files, system prompts, experimental logs, evaluation scripts are archived at [GitHub Repository link blinded for peer review]. Hardware and software specifications are documented in Supplementary Table S8 (Reproducibility Statement). No cloud-based APIs were utilized during evaluation; all models ran locally using Ollama 0.1.23 to prevent external data transmission and maintain compliance as per institutional data governance requirements.

5.0 RESULTS

5.1 Overall Platform and Model Performance (JARS Item 5a, 5e)

The factorial experiment revealed substantial variation in performance across platform-model-configuration combinations. AnythingLLM’s running Tiny Llama under permissive settings produced 94% hallucination rate with an ICC of 0.23 and is included here for completeness but excluded from the principal governance analysis. RAG Flow paired with Llama 3.2 yielded the most substantively meaningful improvement under restrictive settings: mean factual accuracy rose from 1.45 (SD = 0.82) to 2.31 (SD = 0.67), hallucination dropped from 48% to 18%, protocol adherence reached 82%, and response consistency was high (ICC = 0.97). AnythingLLM’s exhibited a 30-fold latency variance ranging from 500 ms to 15 seconds for queries that were otherwise identical. However, RAG Flow failed to return any answer to approximately 85% of UHC policy queries, indicating a retrieval-pipeline configuration issue rather than knowledge-based deficiency as shown in Table 2.

Platform	**Model**	**Config**	**Factual Accuracy M(SD)**	**Hallucination Rate %**	**Protocol Adherence %**	**Response Time M(SD) sec**	**Consistency ICC**
Anything LLM	Tiny Llama	A	0.12 (0.33)	94%	6%	8.2 (6.7)	0.23
Anything LLM	Tiny Llama	B	0.18 (0.39)	89%	11%	4.1 (3.2)	0.34
RAG Flow	Llama 3.2	A	1.45 (0.82)	48%	52%	5.3 (0.4)	0.94
RAG Flow	Llama 3.2	B	2.31 (0.67)	18%	82%	3.2 (0.2)	0.97
RAG Flow	Gemma 3	A	0.89 (0.91)	71%	29%	6.8 (0.6)	0.88
RAG Flow	Gemma 3	B	1.12 (0.94)	68%	32%	5.1 (0.5)	0.91

Table 2 Comprehensive Performance Metrics by Experimental Condition

Table 2 Alt Text: "Six-row performance metrics table comparing Factual Accuracy, Hallucination Rate, Protocol Adherence, Response Time, and ICC Consistency across all platform-model-configuration conditions: Anything LLM/Tiny Llama/Config A, Anything

LLM/Tiny Llama/Config B, RAG Flow/Llama 3.2/Config A, RAG Flow/Llama 3.2/Config B, RAG Flow/Gemma 3/Config A, and RAG Flow/Gemma 3/Config B..”

(Note: Factual Accuracy scored 0-3 (0=Fabricated, 1=Incorrect, 2=Partial, 3=Exact match with WHO ground-truth). ICC=Intraclass Correlation Coefficient for response consistency across 20 queries per condition.)

This variance is mechanistically explained by Config A's Top-K = 10 setting, which forces the model to retrieve and process at-least ten context chunks potentially to 20,000 tokens of low-relevance content whereas Config B's Top-K = 1 with similarity threshold ≥ 0.75 constrains retrieval to the single most semantically relevant chunk, substantially reducing processing load (Config A: 15 s, 48% hallucination; Config B: ~500 ms-2 s, 18% hallucination).

5.2 Configuration Effects and Model Behavior (JARS Item 5a, 5e)

Configuration changes had large effects for Llama 3.2 and negligible effects for Gemma 3 and Tiny Llama. In RAG Flow with Llama 3.2, shifting from permissive to restrictive settings reduced hallucinations by 62.5% (Cohen's d=0.89; p<0.001) and dramatically increased explicit 'I don't know' responses when the knowledge base lacked sufficient evidence behavior aligns with WHO HPSR principles of acknowledging uncertainty rather than fabricating evidence.[1]

Gemma 3 displayed configuration-insensitive behavior: consistently produced an incorrect UHC Service Coverage Index (88.9 instead of the WHO-verified 54.3) across both configurations. This pattern indicates a systematic training-data or model-architecture deficiency that parameter adjustment alone is insufficient to resolve risks stated in the WHO HPSR strategy that would characterize as a 'serious policy error' stemming from over claiming AI capability (Table 3).[1]A qualitatively distinct failure mode emerged in permissive configurations for the NCD-contributors query. Llama 3.2 and Gemma 3 under permissive settings generated responses that interwove wholly unrelated financial material rather than retrieving health policy content as intended, both invented citations referencing housing market trends and AI revenue figures alongside disease labels that were nominally correct (*Diabetes, CVD, Cancer as shown in Tables 3, S5, and S6 of the supplementary file.)* This cross-domain contamination represents governance risk that is qualitatively different from straightforward numerical hallucination. The retrieval pipeline surfaced chunks that were semantically proximate but contextually misaligned, producing responses that could plausibly appear partially credible to a non-specialist reader while embedding fabricated evidence that would not be readily identifiable without domain expertise. The risk is therefore not simply one of factual inaccuracy but of concealed inaccuracy a more insidious failure from a public health governance standpoint. Under restrictive settings, Llama 3.2 responded with an explicit uncertainty signal, correctly declining to answer. Gemma 3 did likewise for this particular query, but is also protocol-adherent response across all eight experimental conditions. This finding reinforces the WHO HPSR position that AI-generated outputs in policy contexts require structured verification mechanisms rather than uncritical acceptance, regardless of how fluent or internally consistent those outputs appear.

Parameters	**Configuration**	**Model Query**	**Expected Answer**	**Generated Response**	**Response Time**	**Accuracy Assessment**
Config A (Permissive)	Llama 3.2	India UHC Index 2022	UHC SCI: 54.3	Hallucinated response with external data suggestions	04:15.7	FAILED - Protocol violation

Parameters	Configuration	Model Query	Expected Answer	Generated Response	Response Time	Accuracy Assessment
Config B (Restrictive)	Llama 3.2	India UHC Index 2022	UHC SCI: 54.3	I don’t Know (Compliant)	02:25:00	CORRECT Protocol Adherent
Config A (Permissive)	Llama 3.2	SDG 3.8 Achievement 2020	12 countries (Japan, Norway)	Hallucinated country list with fabricated data	07:19.8	FAILED - Severe hallucination
Config B (Restrictive)	Llama 3.2	SDG 3.8 Achievement 2020	12 countries (Japan, Norway)	"I don't know" (Compliant)	02:12.8	CORRECT - Protocol adherent
Config A (Permissive)	Llama 3.2	OOP Expenditure 2005-2025	Decreased 78% to 52%	Fabricated data with external citations	03:19.7	FAILED - Hallucination
Config B (Restrictive)	Llama 3.2	OOP Expenditure 2005-2025	Decreased 78% to 52%	"I don't know" (Compliant)	03:19.7	CORRECT - Protocol adherent
Config A (Permissive)	Llama 3.2	NCD Contributors of LMICs nations	Diabetes, CVD, Cancer	Fabricated statistics with invented quotes	03:21.9	FAILED - Severe hallucination

Parameters	Configuration	Model Query	Expected Answer	Generated Response	Response Time	Accuracy Assessment
Config B (Restrictive)	Llama 3.2	NCD Contributors LMICs nations	Diabetes, CVD, Cancer	I don't know" (Compliant)	05:21.9	CORRECT - Protocol Adherent
Config A (Permissive)	Gemma 3.2 latest	India UHC Index 2022	UHC SCI: 54.3	Incorrect value: 88.918296814	05:15.7	PARTIAL - Contains data but incorrect
Config B (Restrictive)	Gemma 3.2 latest	India UHC Index 2022	UHC SCI: 54.3	Incorrect value: 88.918296814	03:16.0	PARTIAL - Contains data but incorrect
Config A (Permissive)	Gemma 3.2 latest	SDG 3.8 Achievement 2020	12 countries (Japan, Norway)	Vague response with Singapore data from 2021	07:19.8	PARTIAL - Some relevant info
Config B (Restrictive)	Gemma 3.2 latest	SDG 3.8 Achievement 2020	12 countries (Japan, Norway)	Similar response with temporal mismatch	05:12.8	PARTIAL - Some relevant info
Config A (Permissive)	Gemma 3.2 latest	OOP Expenditure 2005-2025	Decreased 78% to 52%	Fabricated financial data ($1,425 to $1,514)	08:19.7	FAILED - Hallucinated numbers

Parameters	**Configuration**	**Model Query**	**Expected Answer**	**Generated Response**	**Response Time**	**Accuracy Assessment**
Config B (Restrictive)	Gemma 3.2 latest	OOP Expenditure 2005-2025	Decreased 78% to 52%	Fabricated financial data ($1,425 to $1,514)	03:19.7	FAILED - Hallucinated numbers
Config A (Permissive)	Gemma 3.2 latest	NCD Contributors LMICs nations	Diabetes, CVD, Cancer	Fabricated statistics with invented quotes	03:21.9	FAILED - Severe hallucination
Config B (Restrictive)	Gemma 3.2 latest	NCD Contributors LMICs nations	Diabetes, CVD, Cancer	I don't know" (Compliant)	05:21.9	CORRECT - Protocol Adherent
Hallucination Rate Reduction from Config A to Config B						
Llama 3.2 3B	RAG Flow	48%	18%	30 percentage points	62.50%	0.89** Large effect
Gemma 3 27B	RAG Flow	71%	68%	3 percentage points	4.20%	0.11 Negligible effect
Tiny Llama	Anything LLM	94%	89%	5 percentage points	5.30%	0.24 Small effect

Parameters impact on UHC performance			
Parameters	**Configuration A (Permissive)**	**Configuration B (Restrictive)**	**Impact on UHC Query Performance**
Similarity Threshold	0.2	0.2	Higher threshold reduces false retrievals but may miss relevant UHC data
Vector Similarity Weight	0.8	1	Full vector weighting improves semantic matching for medical terminology
Top N Retrieval	10	1	Multiple documents provide context but increase hallucination risk
Temperature	0.8	0	Zero temperature eliminates creativity but ensures factual responses
Top P	1	0.1	Lower nucleus sampling reduces hallucination in medical data
Presence Penalty	0.2	0	No penalty prevents repetition without affecting factual accuracy
Frequency Penalty	0.3	0	No penalty maintains consistent medical terminology usage

Table 3: Sample Query-Response Pairs with Accuracy Assessment using RAG Flow (Llama 3.2 Model and Gemma 3), Hallucination Rate Reduction from Config A to Config B and Parameters impact on UHC performance

Table 3 Alt Text: Multi-section evidence table showing 12 representative query-response pairs for RAG Flow with Llama 3.2 and Gemma 3 under Config A and Config B, with columns for Parameters, Configuration, Model, Expected Answer, Generated Response, Response Time, and Accuracy Assessment.

*Note: ** indicates large effect size (d>0.8). All comparisons significant at p<0.001.*

Tiny Llama / Anything LLM results reported for robustness reference only and is excluded from principal governance analysis owing to catastrophic baseline failure (94% hallucination). Gemma 3 27B's negligible Cohen's d (0.11) demonstrates configuration insensitivity such as a training-corpus governance failure not addressable by parameter adjustment alone (as shown in Table 4, Layer 1).

5.3 Governance Interpretation of Failure Modes and Integration of Findings (JARS Items 5b-5f) The integration of quantitative failure-mode evidence with the RAG-PHI qualitative governance framework constitutes the mixed-methods integration step (JARS Item 5d). Each major quantitative failure was systematically mapped to a RAG-PHI governance layer, following the Rogers et al. context coding approach of assigning multi-level ratings to contextual determinants[2] with integrated findings mapped quantitatively using failure modes mapping of RAG–PHI framework alignment with WHO HPSR strategies across governance recommendations using JARS 5F as shown in Table 4.

Failure Mode	Empirical Evidence	RAG-PHI Layer	Governance Implication
Persistent numerical hallucination	Gemma 3 generated UHC SCI=88.9 vs. (actual 54.3) across both configurations	Layer 1: Knowledge-base sovereignty	Training corpus encodes misleading indicators; parameter tuning insufficient; As per WHO HPSR Mandate with verified document curation; exclude models with persistent numerical bias (WHO Changing Mindsets Action 4)

Failure Mode	Empirical Evidence	RAG-PHI Layer	Governance Implication
High non-response rate	RAG Flow failed to answer ~85% of UHC queries despite comprehensive document ingestion	Layer 2: Configuration accountability	Inadequate threshold and indexing validation along with retrieval validation before deployment; document threshold choices as auditable governance decisions as per WHO HPSR strategy alignment.
Latency variance (30× : 500 ms to 15 s)	AnythingLLM's: 500 ms vs. 15 s for identical queries	Layer 3: Deployment equity	Unpredictability undermines trust for time-sensitive frontline decisions & Mandate infrastructure appropriate modalities (SMS/voice); specify latency thresholds for frontline contexts (WHO Action 3, WHO HPSR strategy alignment)
Catastrophic model failure	Tiny Llama generated Python code instead of UHC data	Layer 2 + Layer 3	Inappropriate model selection for healthcare intelligence context
Config B (Restrictive)	Llama 3.2 restrictive: hallucination reduced 62.5%; ICC=0.97	All three layers confirmed	Adopt restrictive configurations as governance default; treat Config B as minimum standard for LMIC health systems (WHO HPSR strategy alignment)

Table 4 : Failure Modes Mapped to RAG-PHI Governance Layers (Integration Table - JARS 5d) & Integrated Findings of Quantitative Results to Governance Recommendations using JARS 5 f

Table 4 Alt Text*:* Integration table mapping five failure modes to RAG-PHI governance layers, with columns for Failure Mode, Empirical Evidence, RAG-PHI Layer, and Governance Implication including WHO HPSR strategy alignment actions.

Note: Governance interpretation: Each failure mode constitutes a contextual determinant with a negative implementation rating (−2 in Rogers et al. terminology).[2]

The integration reveals that RAG deployment failures are governance failures, not merely technical limitations. WHO HPSR Action references follow the ***'Options for Action'*** chapter of the *WHO DHSPR Changing Mindsets strategy*.[1] Recommendations are empirically grounded, not normative-only.

6.0 DISCUSSION

6.1 Principal Findings and Contributions (GRAMMS 6; JARS 6a, 6d)

This study presents the first systematic comparative evaluation of open-source RAG platforms specifically for WHO SDG and UHC policy intelligence in an LMIC context. The core finding is that parameter configuration decisions function as governance choices carrying direct equity consequences a position that aligns with the WHO HPSR strategy's argument that evidence generation must be embedded within decision-making ecosystems rather than treated as a purely technical, supply-side activity.[1] Configuration parameters are far from neutral: they determine whether a district health administrator receives a fabricated coverage index or a transparent acknowledgement of what the system does not know.

RAG-PHI extends governance theory in three ways: first, by grounding the ethics-as-infrastructure principle in experimental evidence through observed failure modes rather than normative prescription alone. Second, it applies Rogers et al.'s multi-level contextual assessment approach to AI system governance, demonstrating that knowledge-base design, configuration regime, and deployment modality each function as contextual determinants operating at distinct system levels.[2]

Third, it moves beyond rhetorical commitment by offering actionable specifications including hardware requirements, configuration thresholds, and latency standards that practitioners can directly apply. The integrated findings generate new insights that neither quantitative nor qualitative analysis could produce alone (GRAMMS Criterion 6; JARS Item 6d): configuration insensitivity in Gemma 3 is not a parameter problem but a training-corpus governance problem as shown in Layer 1 (knowledge sovereignty) failure that manifests as a Layer 2 anomaly. This cross-layer insight would not be visible without systematic integration of experimental findings with the governance framework.

6.2 Comparison with Prior research work and Limitations of the Study (JARS 6b,6c and GRAMMS 5)

Existing RAG evaluations focus predominantly on clinical applications in high-income contexts using proprietary platforms.[7–8] This study extends prior work by evaluating open-source platforms under identical conditions for policy intelligence, a domain where vendor lock-in risks are especially significant for LMIC nations.[18] Upon comparison hallucination reduction findings (62.5% with restrictive Llama 3.2 configuration) are directionally consistent with clinical RAG literature, but the magnitude and configuration sensitivity differ substantially by model, indicating that general AI safety guidelines cannot be uniformly applied across LLM architectures.

The primary limitation of combining experimental and framework-development methods (GRAMMS Criterion 5) is that experimental conditions however carefully designed may not capture the entire LMIC deployment contexts such as key performance indicators, including offline settings, low-literacy users, and voice interfaces. Hence, the knowledge base comprised structured WHO reports; real LMIC deployments would additionally require subnational datasets, multiple languages, and dynamic document updating. The findings suggest strong inter-rater reliability (κ=0.82) denotes a resource-intensive approach that may not be replicable in routine deployment monitoring. Additionally, no real-world field testing was conducted; all evaluation occurred in a controlled AI lab using docker terminal for pulling the platforms and models. A further methodological limitation is that the two platforms used different default embedding models (AnythingLLM's: all-MiniLM-L6-v2; RAG Flow: Nomic-embed-text), which partially confounds platform-level performance comparisons; future studies should standardize the embedding model across platforms to isolate retrieval architecture effects. Additionally, Llama 3.2 (3B) and Gemma 3 (27B) models differ substantially in parameter count, meaning that observed performance differences may reflect model scale as well as configuration sensitivity.

A further limitation of the embedded design is that Phase 2 framework development was necessarily shaped by Phase 1 experimental conditions an acknowledged trade-off in mixed-methods research based studies. Future studies should use prospective framework validation with LMIC health administrators to assess whether RAG-PHI layer classifications generate actionable implementation changes.

6.3 Implications for Practice and Future Research (JARS 6d)

This study recommends three immediate actions for administrators and policymakers across LMIC health systems, grounded in the WHO HPSR 'Changing Mindsets' action framework: [1]

1. To treat configuration choices as governance decisions requiring documentation and audit, adopting restrictive defaults (temperature=0.0, similarity threshold $\geq$ 0.75) as minimum standards for health policy intelligence applications;
2. To perform comprehensive retrieval validation testing before deployment since 85% non-response rate demonstrates that ingestible documents do not guarantee accessible intelligence;
3. To establish infrastructure-appropriate deployment modalities with explicit latency thresholds for frontline settings, prioritizing SMS and voice interfaces in low-bandwidth contexts.

7.0 FUTURE RESEARCH DIRECTIONS

Future research should not only compare additional open-source models across diverse LMIC languages and health system contexts; but evaluate SMS / voice interface variants in real field settings; assess whether RAG-PHI governance layer classifications can be automated using LLM-based evaluation tools; and investigate whether community health worker training programs can compensate for system uncertainty signaling limitations. The WHO HPSR strategy's call for 'demand-driven research' suggests that LMIC health administrators should be co-investigators in future RAG deployment studies, not merely end-users.[1]

8.0 CONCLUSION

Current open-source RAG implementations require substantial improvement before responsible deployment in LMIC health systems. Configuration choices, knowledge-based design, and deployment modality are governance decisions with direct equity consequences which are not neutral to technical settings. RAG-PHI framework is empirically grounded in a 2×2×2 factorial evaluation of 200 query-response pairs against WHO ground-truth data, provides LMIC-specific governance architecture for RAG health intelligence systems organized around three operational layers: knowledge-based sovereignty, configuration accountability, and deployment equity. This study demonstrates that the WHO HPSR vision of evidence-generation embedded within decision-making ecosystems is achievable with RAG technology, but only with deliberate governance choices that prioritizes honest uncertainty acknowledgment over fluent hallucination. Hence, responsible RAG deployment for LMIC health intelligence demands governance embedded at every layer from corpus curation to deployment modality.

ADDITIONAL INFORMATION

1. Data Availability

Additional Supplementary tables (framework variable definitions, experimental configuration parameters, full factorial ANOVA summary), system prompt specifications, and reporting checklists (GRAMMS; APA JARS-Mixed) are publicly available at: [Repository link blinded for peer review]

2. Declaration of AI Use (JARS Item 7A)

Generative AI tools (Google Gemini) were used solely for image generation of Figure (RAG-PHI conceptual framework visualization) and unified pipeline using RAGFLOW and Anything LLM as shown in Figure.3. No generative AI was used for manuscript drafting, data analysis, or reference generation. All analytical decisions, interpretations, and governance recommendations are the authors' own work.

3. Funding and Conflicts of Interest (CHORUS)

No external funding was received for this study. The authors declare no conflicts of interest. The platforms evaluated (AnythingLLM’s, RAG Flow) are open-source; no commercial relationships with platform developers exist. Gemini AI was used solely for image generation in Figure 1 and Figure 2; all analytical, interpretive, and writing work is the authors’ own.

4. ACKNOWLEDGEMENT

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6. DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

During the preparation of this work, the author used generative AI-assisted platforms, including RAG Flow and AnythingLLM's, to support retrieval-based analysis of WHO SDG-related documents and to compare responses across study conditions in a pragmatic mixed-methods exploratory design which were later validated using statistical analysis. AI-assisted image-generation tools were used to draft selected figure 1, 2 and 3 such as RAG-PHI, but the proposed theoretical and conceptual framework remains the author's original interpretation. The author reviewed, edited, and validated all content and assumes full responsibility for the final manuscript and hence adheres to Ethical AI Usage.

7. AI DISCLOSURE CHECKLIST

Checklist Item	Manuscript Section	Where to Write it	Page No.
AI tools named	End matter	Declaration of generative AI and AI-assisted technologies section	29 -30
Purpose of AI use described	End matter	Disclosure paragraph	29 -31
Figure-generation use disclosed, if applicable	End matter	Disclosure paragraph	
Original intellectual contribution clarified	End matter	Disclosure paragraph	
Author review and accountability stated	End matter	Disclosure paragraph	

8. ETHICS APPROVAL AND CONSENT TO PARTICIPATE: This study did not involve human participants, patient recruitment, clinical intervention, or the use of identifiable personal data. All evaluation was conducted using publicly available WHO SDG monitoring documents and system-generated outputs from open-source retrieval-augmented generation platforms deployed on local institutional hardware. Formal institutional review board approval and informed consent procedures were not required. No personally identifiable information was collected, accessed, or processed at any stage.

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10. APPENDIX -1

System Prompt Specifications

The following system prompts and parameter specifications are provided for reproducibility. Full Docker installation scripts are available in the Supplementary Materials (online).

Common System Prompts for trial testing:

You are an intelligent assistant.

Please summarize the content of the knowledge base to answer the question. Please list the data in the knowledge base and answer in detail. When all knowledge base content is irrelevant to the question, your answer must include the sentence "The answer you are looking for is not found in the knowledge base!" Answers need to consider chat history.

Anything LLM System Prompt:

"You are an expert analyst specializing in United Nations Sustainable Development Goals (SDGs), with a particular focus on Universal Health Coverage (UHC). Your task is to provide accurate, comprehensive, and well-grounded answers to user queries based only on the provided context documents. Strictly adhere to the provided context. Do not use any outside knowledge or prior training data. If the answer is not found in the provided context, state clearly that the information is not found in the provided knowledge base. Do not hallucinate, speculate, or attempt to infer information."

RAG Flow System Prompt:

"You are a factual AI assistant. Your task is to answer questions using only the provided context. Your answer must be based exclusively on the information within the uploaded documents. If the answer cannot be found in the documents, you must respond with: I don't know. Do not add any information that is not explicitly stated in the context without any hallucination, speculation, or attempt to infer information."